

DAYAPURAM RESIDENTIAL SCHOOL

SHORT NOTE – RATIONAL NUMBERS

- The numbers that can be represented in the form of $\frac{p}{q}$, where $q \neq 0$ is called rational numbers.

Properties of Rational Numbers:

- **CLOSURE PROPERTY:** Rational numbers are closed under addition, subtraction, and multiplication.
- **COMMUTATIVE PROPERTY:** Addition and multiplication are commutative.

$$a + b = b + a \text{ and } a \times b = b \times a$$

- **ASSOCIATIVE PROPERTY:** Rational numbers are associative under addition and multiplication.

$$a + (b + c) = (a + b) + c \text{ and } a \times (b \times c) = (a \times b) \times c$$

- **DISTRIBUTIVE PROPERTY:** Rational numbers are distributive over addition and subtraction.

$$a \times (b + c) = (a \times b) + (a \times c)$$

$$a \times (b - c) = (a \times b) - (a \times c)$$

- **ADDITIVE IDENTITY:** The additive identity is **0**.

$$a + 0 = 0 + a = a$$

- **ADDITIVE INVERSE:** The additive inverse of a number is found by changing the sign of the number.

$$a + (-a) = -a + a = 0$$

Note: The sum of a number and its additive inverse is **0**.

- **MULTIPLICATIVE IDENTITY:** The multiplicative identity is **1**

$$a \times 1 = 1 \times a = a$$

- **MULTIPLICATIVE INVERSE(RECIPROCAL):** The multiplicative inverse can be found by interchanging the numerator and denominator.

$$\text{Reciprocal of } \frac{a}{b} \text{ is } \frac{b}{a}$$

Note: The number **0** has no reciprocal (multiplicative inverse)

Note: The product of a number and its multiplicative inverse is **1**.

QUESTIONS:

- Which of the following is the additive inverse of 0?
a) 0 b) 1 c) -1 d) not defined
- Which of the following is the multiplicative inverse of 0?
a) 0 b) 1 c) -1 d) Not defined

3. The product of a number and its reciprocal is:
 - a) The number itself
 - b) 1
 - c) 0
 - d) none of these
4. The sum of a number and its additive inverse is:
 - a) The number itself
 - b) 1
 - c) 0
 - d) none of these
5. Which of the following is not true?
 - a) Rational numbers are closed under multiplication
 - b) Rational numbers are closed under division
 - c) Rational numbers are closed under addition
 - d) Rational numbers are closed under subtraction
6. Name the property: $\frac{2}{3} \times \frac{4}{5} = \frac{4}{5} \times \frac{2}{3}$
7. Name the property: $\frac{2}{3} + \frac{4}{5} = \frac{4}{5} + \frac{2}{3}$
8. Name the property: $\frac{2}{3} + \left(\frac{4}{5} + \frac{6}{7}\right) = \left(\frac{2}{3} + \frac{4}{5}\right) + \frac{6}{7}$
9. Name the property: $\frac{2}{3} \times \left(\frac{4}{5} + \frac{6}{7}\right) = \left(\frac{2}{3} \times \frac{4}{5}\right) + \left(\frac{2}{3} \times \frac{6}{7}\right)$
10. Verify: $\frac{-4}{9} \times \frac{7}{5} = \frac{7}{5} \times \frac{-4}{9}$.
11. Verify: $\frac{-4}{9} + \frac{7}{5} = \frac{7}{5} + \frac{-4}{9}$
12. Verify: $\frac{2}{3} \times \left(\frac{4}{5} + \frac{6}{7}\right) = \left(\frac{2}{3} \times \frac{4}{5}\right) + \left(\frac{2}{3} \times \frac{6}{7}\right)$
13. Find using the distributive property: $\left(\frac{2}{5} \times \frac{-6}{7}\right) + \left(\frac{1}{5} \times \frac{2}{5}\right)$
14. Find the reciprocal of $\frac{-7}{8} \times \frac{6}{5}$.
15. Find the multiplicative inverse of $\frac{-4}{5} \times \frac{6}{7}$.
16. Using appropriate properties, find: $\frac{5}{6} \times \frac{1}{3} - \frac{1}{2} + \frac{1}{3} \times \frac{2}{3}$